Hyangsuk Min

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Research Profile

I work toward agentic AI systems that remain reliable and trustworthy, from a single generated response to a fully autonomous research loop. I measure reliability through multi-dimensional, cross-lingual evaluation of LLM generation and reliable judgment via multi-agent debate and human–multi-agent collaboration [4–7, 10, 11]. I build it into agents through agentic memory for dialogue systems [8, 12]. Most recently, I have been developing a co-evolving agent harness for autonomous research, in which an agent predicts the outcomes of candidate experiments before running them, selects which to test under a limited budget, and calibrates its predictions against observed results.

Education

- **Korea Advanced Institute of Science and Technology** — *Sep 2024 - Present*
  Ph.D., Industrial and Systems Engineering (advisor: Prof. Hwanjun Song) — Daejeon, South Korea
- **Korea Advanced Institute of Science and Technology** — *Mar 2020 - Aug 2022*
  M.S., Graduate School of Data Science (advisor: Prof. Jae-Gil Lee) — Daejeon, South Korea
- **Kookmin University** — *Mar 2015 - Aug 2019*
  B.S., Bigdata Analysis and Statistics — Seoul, South Korea

Publications

* equal contribution

(12) **Min, H.**, Song, H. (2026). Don't Scroll Back: Missing-Evidence Memory for Streaming Dialogue Summarization. In *Findings of the Association for Computational Linguistics: EMNLP 2026*.

(11) Yun, T*., **Min, H*.**, Lee, Y., Bang, S., Song, H. (2026). Designing Human-Multi-Agent Collaboration for Fact-checking in Crowdsourced Annotation Workflows. (under review)

(10) Ban, M*., Choi, J*., **Min, H*.**, Kim, NH., Kim, M., Lee, J., Song, H. (2026). Completing Missing Annotation: Multi-Agent Debate for Accurate and Scalable Relevant Assessment for IR Benchmarks. In Proc. *the 14th International Conference on Learning Representations (ICLR)*.

(9) Ban, M., Choi, J., **Min, H.**, Yun, T., Oh, J., Song, H. (2026). A Benchmark Dataset for Retrieval-Augmented Generation Considering Performance Alignment between Retrieval and Generation. *KIISE Transactions on Computing Practices*, 32(4).

(8) **Min, H.**, Song, H. (2025). Streaming Dialogue Summarization: A Memory-Augmented Framework for Summarization and Evaluation. In *Korea Software Congress (KSC)*.

(7) Ban, M*., Choi, J*., **Min, H*.**, Kim, NH., Bang, S., Song, H. (2025). BRIDGE: Toward Faithful RAG Benchmarking via Retrieval-Generation Alignment. In Proc. *the 34th ACM International Conference on Information and Knowledge Management Workshop (CIKM Workshop)*.

(6) Lee, Y*., Deng, J*., Kim, NH., **Min, H.**, Yun, T., Ban, M., Kim, Y., Song, H. (2025). Towards a Holistic and Automated Evaluation Framework for Multi-Level Comprehension of LLMs in Book-Length Contexts. In Proc. *the 2025 Conf. on Empirical Methods in Natural Language Processing (EMNLP)*.

(5) Yun, T., Oh, J., **Min, H.**, Lee, Y., Bang, J., Cai, J., Song, H. (2025). ReFeed: Multi-dimensional Summarization Refinement with Reflective Reasoning on Feedback. In Proc. *Conf. on Language Modeling (COLM)*.

(4) **Min, H*.**, Lee, Y*., Ban, M., Deng, J., Kim, NH., Yun, T., Su, H., Cai, J., Song, H. (2025). Towards Multi-dimensional Evaluation of LLM Summarization across Domains and Languages. In Proc. *the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*.

(3) **Min, H.**, Lee, J. (2023). Temporal Convolutional Network-Based Time-Series Segmentation. In Proc. *2023 IEEE Int'l Conf. on Big Data and Smart Computing (BigComp)*.

(2) Kim, D., **Min, H.**, Nam, Y., Song, H., Yoon, S., Kim, M., Lee, J. (2022). COVID-EENet: Predicting Fine-Grained Impact of COVID-19 on Local Economies. In Proc. *36th AAAI Conf. on Artificial Intelligence (AAAI)*.

(1) Kim, M., Kang, J., Kim, D., Song, H., **Min, H.**, Nam, Y., Park, D., Lee, J. (2020). Hi-COVIDNet: Deep Learning Approach to Predict Inbound COVID-19 Patients and Case Study in South Korea. In Proc. *26th ACM SIGKDD Int'l Conf. on Knowledge Discovery and Data Mining (KDD)*.

Professional Experience

- **M2MCx, SW Campus, HL Mando** *Oct 2022 - Aug 2024*
  AI Research Engineer Pangyo, South Korea
  - Built auto-ML training, evaluation, and deployment pipelines for smart factory and vehicle diagnostics systems.
  - Developed on-premise MLOps infrastructure and CI/CD workflows for deep learning models using sensor and image data in production environments.
  - Designed synthetic data generation pipelines for vision inspection and uncharted driving scenarios under limited-data and distribution-shift settings.
  - Applied prompt engineering and domain adaptation to LLM-based OEM requirement classification and mixed-format document analysis.
  - Worked across the full ML lifecycle, including data analysis, model development, evaluation, deployment, and production monitoring.
- **Physical Design, Uniquify** *Aug 2018 - Jun 2019*
  ASIC Engineer Intern San Jose, CA, USA
  - Worked with overseas engineering teams on physical design, PnR, and engineering change order workflows.
  - Developed Perl scripts to summarize timing reports and identify design violations.

Honors and Awards

- **Graduated Cum Laude** *Aug 2019*
  Kookmin University
  - GPA: 4.4/4.5
- **Merit Scholarship** *Sep 2015 - Mar 2019*
  Kookmin University
  - Awarded merit scholarships totaling approximately KRW 17M ($12,000) across 7 semesters for academic excellence.
  - 2 full, 5 partial scholarships.

Certifications

- **ADP: Advanced Data Analytics Professional** *Jul 2021*
- **SQLD: SQL Developer** *Apr 2018*

Patent

- **Method and Apparatus for Predicting Fine-Grained Impact of COVID-19 on Local Economies** *Oct 19, 2022*
  Registration Number: 1020220134903 South Korea

Skills

- **Programming Languages:** Python (PyTorch), R
- **Web Technologies:** CSS, HTML
- **Database Technologies:** SQL
- **Languages:** Korean (Native), English (Fluent)

Activities

- **Reviewer:** ICML, ACL, EMNLP, NeurIPS, ICLR, Neural Networks

Teaching Experience

- **LLM Fine-tuning with HuggingFace (Assistant instructor, SKT):** Nov, 2023
- **Data Structures and Analysis (IE260 lecture TA, KAIST):** Spring, 2025
- **Programming for Data Science (DS801 lecture TA, KAIST):** Spring, 2026
- **Information Technology for IE (IE261 lecture TA, KAIST):** Fall, 2026

Updated as of October 30, 2026